

BRING YOUR IMAGINATION TO SHAPE WITH 3D PRINTING

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Create a model in any 3D modelling software (Credit: www.3dhubs.com)



3D-print the model using a 3D printer (Credit: www.3dhubs.com)

3D printing is the process of giving life to our imagination. Solid 3D objects are printed layer by layer using a computer and a digital file. 3D printing does not involve cutting/hollowing out a piece of plastic by traditional machines. This enables us to produce desired shapes using lesser material.

The 3D printing industry has evolved worldwide to change the way we live, work or play, and has transformed almost every major industry. 3D printing enables us to design custom models, fast prototyping and complex geometry at a lower cost.

People from different professions are using different types of printers for different purposes, like students for models and design solutions, car manufacturers for parts and prototypes, doctors for prosthetics and medical devices like hearing aids, dentists for braces and retainers, aircraft manufacturers for lightweight parts, prop makers for cinema, product designers for problem-solving designs, architects for scale models, design entrepreneurs, engineers, drone enthusiasts, shoe manufacturers, consumer product makers and so on.

Applications in art and education

3D printers give life to our imagination. Artists are able to bring their ideas into reality via digital models. This technology unleashes a new medium of creativity for adults and children alike, who can now create functional products, visualise the concepts better and learn via hands-on experience.

In the next few years, the number of students in elementary, middle and high schools incorporating 3D printing into their curriculum will likely increase. It is better for students to learn by practice than through theory. This will help them in shaping their future better because when they complete their education, almost every industry will be using 3D printing technology.

3D printing is already being used as a tool to teach students about engineering and design. A number of renowned universities all over the world, such as University of Melbourne, have included their use in the curriculum. To make students future ready, it seems like a good way to introduce the principles of 3D modelling, design, manufacturing and sustainability in their curriculum.

Every 3D print starts with a digital 3D design file, like a blueprint, for a physical object. Time taken to complete a print depends on size of the object and speed of